Ecommerce Sales & Profitability Analysis

End-to-End Business Dashboard using Power BI, SQL, and Python

Business Objective

Analyze ecommerce sales data to identify revenue drivers, profitability risks, discount effectiveness, customer value, and shipping cost impact to support data-driven decision making.

Key Questions Answered

- Which products drive profit vs loss?
- Are discounts improving revenue or hurting margins?
- Which customers are most profitable?
- How does shipping impact profitability?

Dashboard Sections

Executive Sales & Profit Overview

Product Performance & Discount Insights

Discount Impact & Profitability Analysis

Customer Performance & Profitability Analysis

Shipping Performance & Profitability Analysis

Data Source: Ecommerce Sales Dataset
Time Period: 2011 – 2014

Tools Used : - (EDA, Cleaning) (Joins, CTEs, Aggregations) (DAX, Modeling)

Executive Sales & Profit Overview

Ecommerce Performance Summary

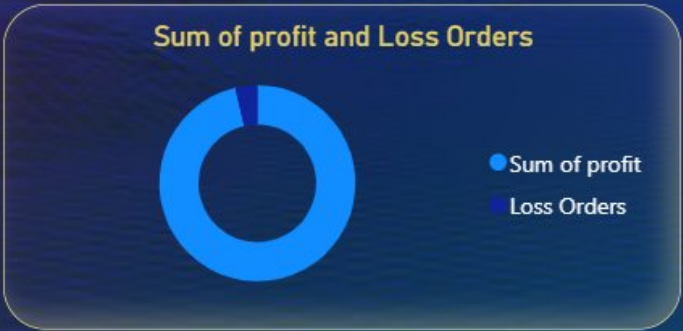
28.44K	1.38K	0.05	159	0.33
Total Sales	Total Profit	Profit Margin %	Total Orders	Loss % Orders

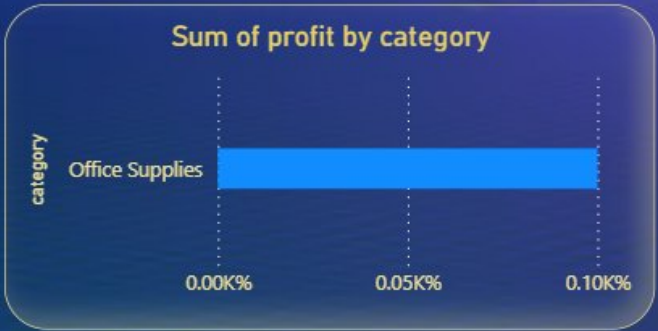
category
- Furniture
- Office Supplies
- Technology

region
- Central
- East
- South
- West

order_year
- 2011
- 2012
- 2013
- 2014

segment
- Consumer
- Corporate
- Home Office





Product Performance & Discount Insights

Top & Bottom Products • Category-wise Sales Impact

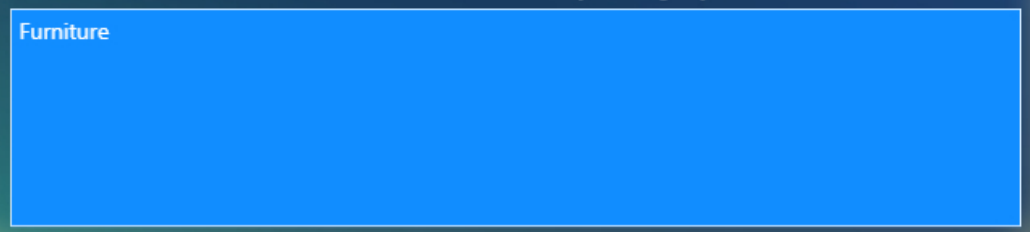
HON 5400 Series...
Top selling product

Hon Deluxe Fabric...
Top profitable product

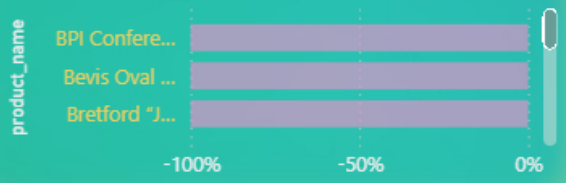
Anderson Hickey...
Worst Product

Top 10 Products by Sales

Sales Contribution by Category



Bottom 10 product by profit



0.17
Avg Discount %

Product Performance Summary

product_name
12-1/2 Diameter Round Wall Clock
12-1/2 Diameter Round Wall Clock
24 Hour Round Wall Clock
Total

Filter by Category & Sub-Category

- Furniture — Bookcases
- Office Sup... — Appliances
- Technology — Accessories

Discount Impact & Profitability Analysis

Understanding how discounts affect profit margins and revenue loss

0.17
Avg Discount %

0.20
Revenue Loss %

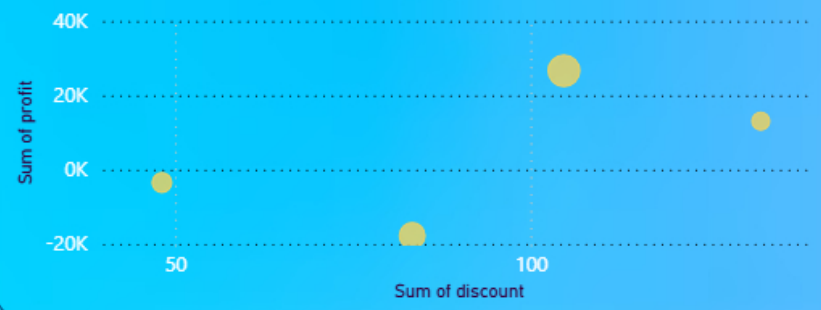
0.02
Profit Margin (Actual)

0.02
Profit Margin (Potential)

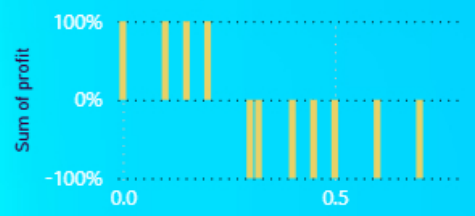
Product & Customer Filters

- **Furniture** — Bookcases
- Office Supplies — Appliances
- Technology — Accessories

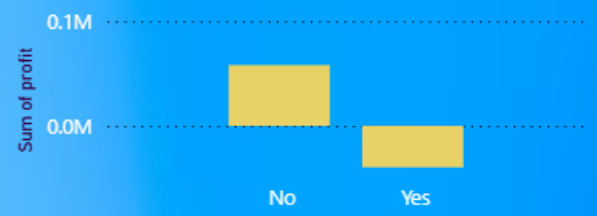
Discount vs Profit by Sub-Category



Sum of profit by discount



Sum of profit by discount_flag



Customer Performance & Profitability Analysis

Analyzing customer value, loyalty, and profit contribution by segment

148
Customer base size

2.90K
Revenue quality

407.50
Customer profitability

147
Customer loyalty

Top 10 Customers by Sales

customer_id	Sum of sales	Sum of profit
TA-21385	14,595.62	4,703.79
DR-12940	8,350.87	2,869.08
AB-10060	7,755.62	2,054.59
ME-17320	10,663.73	1,859.47
Total	**1,05,287.34**	**16,026.95**

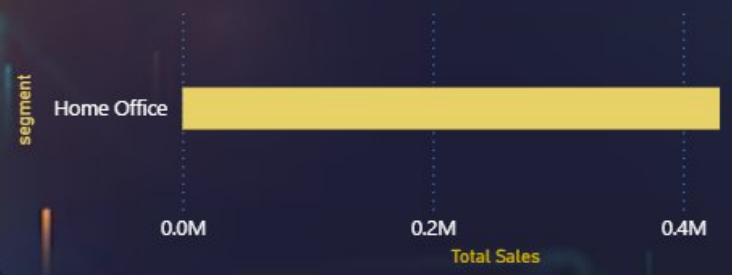
First region and First profit_flag by segment

Consumer	Corporate	Home Office
Central	Central	Central

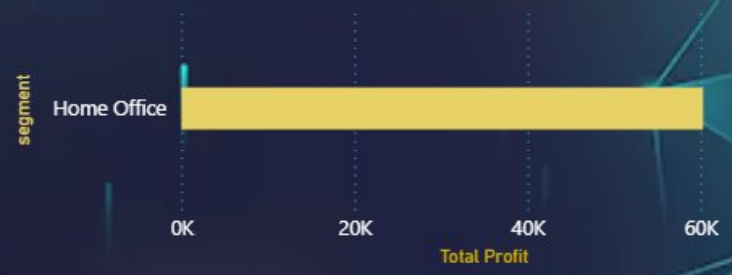
Top 10 Loss-Making Customers

customer_id	Total Profit	Total Sales
AB-10060	2,054.59	7,755.62
AD-10180	1,869.93	6,106.88
DR-12940	2,869.08	8,350.87
ME-17320	1,859.47	10,663.73
Total	**23,729.40**	**77,221.79**

Total Sales by Customer Segment



Total Profit by Customer Segment



Shipping Performance & Profitability Analysis

Evaluating delivery speed, delays, and profit impact across ship modes

5.01
Avg Shipping Days

0.61
Orders with Delay %

Standard Class
Fastest Ship Mode

Standard Class
Most Profitable Ship Mode

Average Shipping Days by Ship Mode

Shipping Days vs Profit by Ship Mode

Total Profit by Ship Mode

Orders with High Shipping Delay

order_id	customer_id	ship_mode	Sum of ship_days	Sum of profit
CA-2013-108196	CS-12505	Standard Class	21	-6,892.37
CA-2014-134845	SR-20425	Standard Class	30	-3,424.35
US-2014-122714	HG-14965	Standard Class	6	-2,929.48
US-2014-120390	TH-21550	Standard Class	7	-1,306.55
CA-2012-146262	VW-21775	Standard Class	35	-1,196.06
Total			**11553**	**53,589.03**

First category and First region by ship_mode

First Class
Furniture

Same Day
Furniture

Second Class
Furniture

Standard Class
Furniture